A PRODUCT OF VALUATION RINGS HAS WEAK DIMENSION AT MOST ONE, AND ITS FRACTION FIELD PRODUCT IS A LOCALIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author's closing sentence reads 'The final statement is true because K = S^{-1}A where S = prod (A_j minus {0})'; it is delivered verbatim. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The weak-dimension-one criterion: it suffices that every finitely generated ideal be flat. invoked at proof lines 30657-30659. Reduces the global statement to a statement about finitely generated ideals.
- Valuation rings have principal ideals, and the image of a finitely generated ideal in a product factor is its tensor product with that factor. used at proof lines 30660-30668. Produces the single generator f of I and reduces the flatness question to the ideal (e).
- A principal ideal generated by an idempotent is a direct summand, hence flat. used at proof lines 30673-30674. Is the flatness conclusion that gives weak dimension at most one.

Source reference. Original lemma 090Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11418–11418. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 059J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6817–6817. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 30575: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30575–30575. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let J be a set. For each $j \in J$ let A_j be a valuation ring with fraction field K_j. Set $A = \prod A_j$ and $K = \prod K_j$. Then A has weak dimension at most 1 and $A \to K$ is a localization.*

Proof. Let $I \subset A$ be a finitely generated ideal. By Lemma 30575 it suffices to show that I is a flat A-module. Let $I_j \subset A_j$ be the image of I. Observe that $I_j = I \otimes_A A_j$, hence $I \to \prod I_j$ is surjective by Algebra, Proposition 059J. Thus $I = \prod I_j$. Since A_j is a valuation ring, the ideal I_j is generated by a single element (Algebra, Lemma 090Q). Say $I_j = (f_j)$. Then I is generated by the element $f = (f_j)$. Let $e \in A$ be the idempotent which has a 0 or 1 in A_j depending on whether f_j is 0 or not. Then $f = ge$ for some nonzerodivisor $g \in A$: take $g = (g_j)$ with $g_j = 1$ if $f_j = 0$ and $g_j = f_j$ else. Thus $I \cong (e)$ as a module. We conclude I is flat as (e) is a direct summand of A. The final statement is true because $K = S^{-1}A$ where $S = \prod(A_j \setminus \{0\})$. □